

Overall Failure Analysis of General Rocket Motor Static Testing

The Solid Propulsion Team

July 3, 2026

1 Introduction

Static testing exists to find problems before flight does. When a test deviates from the expected pressure or thrust trace, or when hardware comes back damaged, the immediate temptation is to name a cause from intuition and move on. This document exists to resist that temptation: it sets out a standardized failure analysis framework, built explicitly on the scientific method, for investigating any anomaly observed during static testing of a solid rocket motor.

The framework is intentionally general. It applies whether the anomaly is an ignition failure, an unexpected pressure spike, a structural failure of the case or nozzle, a lower-than-expected total impulse, or any other deviation from the nominal, predicted behaviour of the motor under test. Treating each of these as a hypothesis-testing problem, rather than a single guess to be confirmed, reduces the risk of two specific failure modes of the analysis process itself: converging on a plausible-sounding but wrong cause, and implementing a corrective action that does not actually address the failure’s origin.

This version grounds that general framework in the specific motor it is meant to be used on: a KNSB-propellant motor in an aluminum 6063-T5 case. Section 2 sets out the propellant and casing theory that the rest of the document depends on, so that “what nominal looks like” for this motor is stated explicitly before any hypothesis is generated against it.

2 Theoretical Background

A hypothesis cannot be predicted, tested, or eliminated without first knowing what the motor is supposed to do. This section establishes that baseline for the two subsystems most directly implicated in the catalog of Section 4: the KNSB propellant grain itself (Section 2.1), and the aluminum 6063-T5 outer casing that contains it (Section 2.2).

2.1 Propellant Theory: KNSB Chemistry and Internal Ballistics

The propellant-side theory below is reproduced verbatim from the team’s internal reference document, *Theory of KNSB*, so that this framework and the propellant theory it depends on remain traceable to a single source rather than a paraphrase of it.

KNSB stands for Potassium Nitrate and Sorbitol (often referred to as “Rocket Candy”).

It is a widely used composite solid propellant in amateur and experimental rocketry. The theory behind its function, chemistry, and ballistics mirrors standard solid propulsion but relies on a unique sugar-alcohol fuel.

The Core Chemical Reaction

Like traditional black powder, KNSB is a heterogeneous mixture that does not rely on atmospheric oxygen to burn. It contains a solid oxidizer and a solid organic fuel melted together.

- **Oxidizer (KNO_3):** Potassium nitrate, which provides the oxygen required for combustion.
- **Fuel ($\text{C}_6\text{H}_{14}\text{O}_6$):** Sorbitol, a sugar alcohol that serves as the combustible binder and carbon/hydrogen source.

When ignited, these components undergo thermochemical conversion at a standard operating pressure of sixty-eight atmospheres (one thousand psi).

The burn rate of KNSB is governed by Saint-Robert's Law (Vielle's Law), where the burn rate is primarily dictated by combustion chamber pressure:

$$\text{Burn Rate} = \text{Burn Rate Coefficient} \times (\text{Combustion Chamber Pressure})^{\text{Pressure Exponent}}$$

or

$$r = aP_c^n$$

where:

- r = burn rate
- a = burn rate coefficient
- P_c = combustion chamber pressure
- n = pressure exponent

Propellant Grain Geometry

Propellant is generally cast as a "grain" with a center-perforated core. The theory of Internal Ballistics states that the thrust curve of a motor is a direct function of the burning surface area. By shaping the core (e.g., a simple cylindrical tube, a rod-and-tube, or a star pattern), rocketeers control the surface area over time, achieving progressive (thrust increases), regressive (thrust decreases), or neutral (thrust stays flat) flight profiles.

Geometry and KN Particle Size

In KNSB rocketry, grain geometry dictates the shape of the thrust curve, while potassium nitrate (KNO_3) particle size dictates the speed of the burn. Balancing both properties ensures a rocket motor generates stable, predictable pressure without exploding.

Part 1: Grain Geometry & The Kn Ratio

Grain geometry refers to the physical shape of the solid propellant segment. In solid rocketry, a critical value known as Kn is used to design the motor:

$Kn = \text{Burning Propellant Surface Area} \div \text{Nozzle Throat Area}$

or

$$K_n = A_b/A_t$$

where:

- A_b = area of the burning propellant surface
- A_t = area of the nozzle throat

Because the nozzle hole (A_t) stays a fixed size, changes to the burning surface area (A_b) change the internal pressure. Rocket designers use three main geometric profiles.

BATES Grain Configuration (Most Common)

- **Design:** Multiple short, hollow cylinders (segments) placed end-to-end. Both the inner core and the flat circular ends of each cylinder are exposed to the flame.

Part 2: Potassium Nitrate (KNO_3) Particle Size

The physical size of the KNO_3 oxidizer crystals drastically alters how efficiently and quickly the fuel burns.

1. Finely milling potassium nitrate (KNO_3) increases the burn rate by maximizing the surface area of the oxidizer, allowing for more contact points with the fuel. Furthermore, small particles shorten the diffusion distance for gas molecules and bring the flame zone closer to the propellant, causing rapid heat transfer and faster ignition.
2. When potassium nitrate (KNO_3) particles are too large, the fuel and oxidizer cannot mix well at a microscopic level. This creates a sluggish chemical reaction that leads to three main problems:
 - Slow flame rate
 - Chuffing (pulsing thrust)
 - Pressure drop and flame out

Major Particle Size Finding

For maximum combustion efficiency, KN particle size should follow a bimodal distribution centered between fifty and one hundred microns.

2.2 Casing Theory: Aluminum 6063-T5 Pressure-Vessel Behavior

The outer casing's job is mechanically simple to state and easy to get wrong in practice: contain the chamber pressure generated by the KNSB burn (Section 2.1) for the full duration of the burn, without yielding, without losing its grip on the closures, and without conducting enough heat through the wall to soften the metal before the motor tails off. For a cylindrical motor case, the standard starting model is the thin-walled pressure vessel: provided the wall thickness t is small relative to the case's mean radius r (conventionally $t/r < 0.1$), the wall can be treated as carrying two uniform stress components rather than a stress field that varies through the thickness.

Hoop and longitudinal stress. For internal pressure P_c , mean radius r , and wall thickness t :

$$\sigma_{\text{hoop}} = \frac{P_c r}{t}, \quad \sigma_{\text{long}} = \frac{P_c r}{2t}$$

The hoop stress is twice the longitudinal stress and is therefore the one that governs case sizing; a case that splits open under pure internal pressure, rather than failing at a closure, fastener, or weld, will typically do so along its length, consistent with a hoop-stress-driven failure.

Factor of safety. The case’s structural margin is the ratio of the material’s yield strength to the operating hoop stress:

$$FS = \frac{\sigma_y}{\sigma_{\text{hoop}}}$$

A motor that has flown or fired safely many times on a given case design is not proof that FS is comfortably above 1. KN-class motors show test-to-test pressure variation driven by Kn drift, grain density variation, or KN particle-size variation (Section 2.1); a case run close to its design FS is at risk of an otherwise-unremarkable pressure excursion finding the margin on the test that happens to run hot.

Material: 6063-T5. Table 1 gives representative mechanical properties for 6063-series aluminum, including the T5 temper specified for this casing. One caveat is necessary and should be treated as a standing item in any case-related hypothesis: T5 is not one of 6063’s standard, widely-characterized tempers. The alloy is most commonly supplied and documented as O (annealed), T4, T6, or T651; T5 (cooled from an elevated-temperature forming operation without a separate solution heat treatment) is far more commonly associated with the thinner-walled 6063 alloy, and 6063-T5 does not have the kind of standard reference datasheet that T6 and T4 do. Published T5 figures for 6063 vary noticeably between suppliers as a result. For a pressure-retaining structural member, this is itself a risk factor worth tracking, not just a footnote: the actual yield strength of the specific stock used should be confirmed against its mill certification, or by coupon test, before being used as a design margin, rather than designed against a single handbook value taken on faith.

Thermal softening as a distinct failure path. A case can fail well below its room-temperature yield strength if the wall is heated significantly by the combustion gas — through liner/insulation degradation, a localized burn-through, or simple under-design of the insulation thickness for the burn duration. Aluminum alloys lose yield strength steeply with temperature well below their $\sim 585^\circ\text{C}$ melting point; a section of case that has lost its insulating liner can soften and rupture under otherwise-nominal chamber pressure. This is why “localized case scorching or hot spot” in Table 3 should be treated as a thermal hypothesis (liner/insulation defect) first and a structural hypothesis (undersized FS) second, and why the post-test evidence standard in Section 3.5 calls for recovered-hardware photographs and residual mass: a thinned, discolored wall section is the signature that distinguishes thermal softening from a structural overload, where crack propagation from a stress riser would instead be expected.

2.3 Linking Theory to Hypothesis Generation

The two subsystem models above are not background reading; they are what makes the “Material” category in Section 3.3 concrete for this motor specifically, rather than a generic checklist item. A

Property	Aluminium 6063-T5	Stainless Steel G316	Units
Outer Diameter	100	101.6	mm
Inner Diameter	94	97.6	mm
Grain Diameter	88	91.6	mm
Peak Pressure	4.26	4.63	mm
Yield strength, σ_y	110	205	MPa
Ultimate tensile strength	150	515	MPa
Young's modulus, E	70	200	GPa
Min. Required t, (Sf = 1.5)	2.73	1.7	mm
Safety Margin vs. MEOP	$\sim 2.25x$	$\sim 4.56x$	-
Simulated Impulse	8,114.60	9,135.88	Ns
Simulated Thrust	2,458	2,357	N
Density	~ 2.8	~ 2.70	g/cm ³

Table 1: Representative mechanical properties for Al 6063-T5 and Stainless Steel G316. Both values are drawn from standard ASTM/MatWeb datasheet as submitted in the GitHub repository under casing mechanical properties.

grain-side hypothesis (void, particle size, batch deviation) predicts a transient, localized pressure signature per Section 2.1; a casing-side hypothesis (marginal FS , thermal softening, off-spec temper) predicts a sustained pressure rise to a hoop-stress-governed failure, or a thermally-thinned wall section, per Section 2.2. Stating explicitly which model a given hypothesis comes from is what allows Section 3.4 to specify a falsifiable, subsystem-appropriate prediction rather than a generic one.

3 Methodology: The Scientific Method Applied to Failure Analysis

Each stage of the classical scientific method has a direct counterpart in post-test failure analysis, applied here against the propellant and casing theory of Section 2. Table 2 gives this mapping, and Figure 1 shows it as the closed loop it actually is: a conclusion is not the end of the process, it is the input to a corrective action whose effectiveness must itself be verified by re-test.

3.1 Observation and Evidence Collection

Evidence quality is the ceiling on everything that follows; no amount of careful reasoning recovers a conclusion from data that was never collected. As a minimum, every static test — not only ones that fail — should produce:

- Synchronized pressure and thrust DAQ traces, sampled fast enough to resolve transients, not only steady-state levels.
- Video of the full test, including a pre-ignition reference frame and the recovery/post-fire condition.

Scientific method stage	Failure analysis equivalent	Output
Observation	Evidence and data collection	Raw dataset: DAQ traces, video, photographs, logs
Question	Anomaly / problem statement	A precise, falsifiable description of the deviation
Hypothesis	Candidate failure modes	A ranked list of plausible causes
Prediction	Expected evidence signature	What each hypothesis implies should, and should not, be observed
Experiment	Verification / targeted testing	Evidence gathered specifically to test the predictions
Analysis	Elimination and comparison	Hypotheses ruled in or out against the evidence
Conclusion	Root cause determination	The most-supported cause(s), stated with its confidence and caveats

Table 2: Mapping between the scientific method and failure analysis stages.

- Pre-test photographs and mass/dimensional records of the grain and assembled motor.
- Post-test photographs, recovered hardware mass, and any slag/residue retained.
- An environmental log: ambient temperature, humidity, and any other condition that could plausibly affect the propellant or hardware.
- The grain’s production batch record (which procedure, which batch, which operator) and its pre-test QA checklist.
- Operator/witness notes, including anything that seemed unusual even if it was not, at the time, considered significant.

Collecting this for every test, not just retroactively after something goes wrong, is what makes a comparison against nominal behaviour possible at all.

3.2 Problem Statement (Anomaly Definition)

Before any cause is proposed, the deviation itself must be stated precisely enough to be falsifiable. A useful problem statement specifies what was observed, when it occurred relative to ignition (t_0), and how it differs from the expected or nominal trace, in magnitude as well as kind. “The motor failed” is not a problem statement; “chamber pressure rose linearly to 140% of the predicted steady-state value over 0.4 s beginning at $t = 1.2$ s, before the case vented at the forward closure” is.

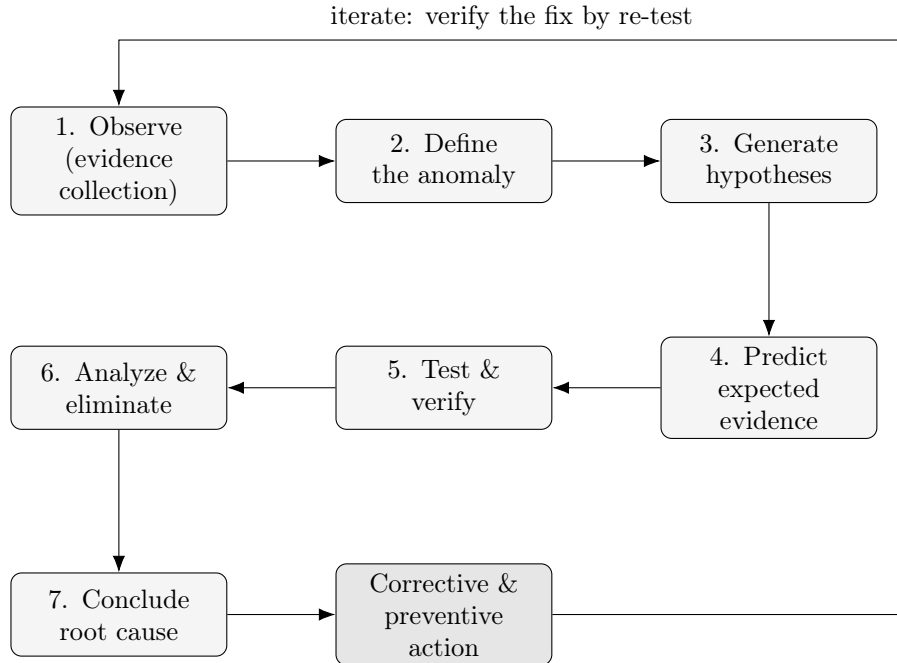


Figure 1: The failure analysis cycle. A conclusion is not a terminal state; it feeds a corrective/preventive action whose effect must itself be confirmed by observation on a subsequent test.

3.3 Hypothesis Generation

Generate candidate failure modes *before* looking too hard for confirming evidence of any one of them; anchoring on the first plausible story is the most common way this process goes wrong. A standard brainstorming structure (adapted from the Ishikawa/fishbone “6 Ms”) is to consider, for the specific anomaly defined above, whether it could plausibly originate from:

- **Material** — propellant grain (voids, microcracks, contamination, incorrect formulation or mix ratio per Section 2.1), liner/insulation, case material (factor of safety and temper verification per Section 2.2), seals.
- **Method** — a deviation from the documented production, assembly, or test procedure.
- **Machine** — test stand, instrumentation, ignition hardware, nozzle.
- **Man** — operator error in assembly, instrumentation setup, or test execution.
- **Measurement** — sensor fault, calibration drift, or DAQ error producing an artifact that looks like a physical anomaly but is not.
- **Environment** — temperature, humidity, or storage conditions affecting the propellant or hardware prior to test.

Section 4 gives a reference catalog of common static-test anomalies and the candidate causes typically associated with each, as a starting point for this stage rather than a substitute for it.

3.4 Predictions

For every hypothesis carried forward, state explicitly what evidence would be expected if it were true — and, just as importantly, what would *not* be expected. A hypothesis that cannot in principle be contradicted by any evidence is not useful here. For example, if the hypothesis is “a grain void reached the burn front,” the prediction is a localized, transient pressure/thrust spike with no corresponding change in measured nozzle throat condition, and sister grains from the same batch should show below-nominal density on the porosity check (cross-referencing the verification protocol in the grain curing procedure). If instead the trace shows a slow, sustained pressure rise with no localized spike, that prediction fails, and the void hypothesis should be weighed down accordingly rather than retained by default.

3.5 Testing and Verification

Where the existing evidence does not clearly confirm or rule out a hypothesis, gather more, specifically targeted at the prediction in question, rather than re-examining the same dataset for a different interpretation. Useful techniques include:

- Destructive sectioning of sister grains from the same production batch, to check directly for the void/microcrack signature predicted above.
- Correlating the DAQ anomaly’s onset timestamp against the corresponding video frame, to check for a visible event (plume change, flash, debris) coincident with the trace deviation.
- Bench-testing isolated subsystems implicated by a hypothesis (e.g. the igniter circuit, a sensor against a reference) independent of a full motor test.
- Fault tree construction, where multiple hypotheses interact (e.g. a void *and* a marginal case safety factor), to make the logical combination of contributing causes explicit rather than implicit.

3.6 Analysis and Elimination

Work through the hypothesis list systematically against the evidence and predictions above, ruling out those that are contradicted and noting the strength of support for those that are not. It is normal, and should be reported honestly, for more than one contributing factor to remain supported; solid rocket motor failures are frequently the product of two marginal conditions combining (e.g. a slightly under-strength case *and* a slightly larger-than-typical void) rather than one dominant cause acting alone.

3.7 Conclusion and Corrective/Preventive Action

State the most-supported root cause (or causes), explicitly noting the confidence level and any hypotheses that could not be fully ruled out with available evidence. From this, propose a corrective action that addresses the root cause directly — not merely the symptom — and a verification plan (typically, a re-test under controlled conditions, per Section 3.5) to confirm that the action actually resolves the issue before it is adopted as standard practice. This closes the loop in Figure 1: the corrective action’s success or failure is itself a new observation, feeding the next iteration of this same process.

4 Failure Mode Reference Catalog

Observed anomaly	Candidate causes to consider
No ignition / delayed ignition	Igniter fault; propellant surface contamination or moisture; ignition circuit or connection fault; insufficient igniter energy for the grain's exposed surface
Pressure/thrust spike mid-burn	Grain void or microcrack reaching the burn front; localized case defect; transient nozzle throat obstruction
Rapid pressure rise to case failure (CATO)	Excessive burning area from crack propagation; nozzle blockage; case structural defect or marginal safety factor (Section 2.2)
Lower-than-expected total impulse	Incomplete combustion; propellant mass measurement error; nozzle throat erosion/enlargement; slag retention in the case
Erratic or noisy thrust trace with no visible anomaly on video	Test stand mechanical resonance or mounting issue; instrumentation/DAQ noise; genuine but benign combustion instability
Localized case scorching or hot spot	Insulation/liner defect at that location; localized void burn-through reaching the case wall; thermal softening of the 6063-T5 wall (Section 2.2)
Asymmetric or uneven burn pattern on recovered hardware	Non-uniform grain density (incomplete mixing or degassing); off-center core/mandrel during casting

Table 3: Reference catalog of common static-test anomalies and candidate causes. This is a starting point for hypothesis generation (Section 3.3), not a substitute for evidence-based verification.

5 Evidence and Instrumentation Standard

To make the process above possible after the fact, the following should be treated as a standing requirement for every static test, independent of whether an anomaly is anticipated:

1. Pressure and thrust DAQ active and logging from before ignition command through full tail-off and cool-down.
2. Video recording covering the full motor and test stand, started before ignition.
3. Pre-test record: grain batch/lot ID, production procedure revision used, mass, dimensions, and QA checklist sign-off.
4. Post-test record: recovered hardware condition, photographs, residual mass, and any deviation from the test plan noted by the operator at the time, not reconstructed afterward from memory.

6 Failure Analysis Report Template

6.1 Static Test 1

Test ID / date	ST005/03/2025
Motor Iteration	M2410
Personnel	All respective teams involved
Anomaly (problem statement)	Semi-immediate explosion with casing failure at the $\frac{2}{3}$ point.
Timeline of events	07.03.2025

6.1.1 Grain, Igniter and Casing Parameters

Grain Nature	KNSB
No of grains used	5
Length	146mm
External Diameter, D ,	86mm
Internal Diameter, d ,	33mm
Port-to-Throat ratio	2.00
Lift-to-Drag ratio	8.43

Igniter type	KNSB + Fe_2O_3
Igniter Length	≈ 30 mm
Length	≈ 140 mm

Casing Length	850mm
Thickness	3mm
External Diameter	100mm
Safety Factor	1.648

6.1.2 Failure Analysis Results

Hypothesis	Predicted evidence	Observed evidence	Supported?
Back Burn	Porous Liner Material and unsuccessful Inhibition	Deep deposition of soot that passed the liner	Yes
Uncharacterised Motor utilisation	Burn rate was higher than theorized values	Longer burn time than calculated/simulated.	Yes
Irregular Mass Flux behavior	Irregular behavior by independent grains that affect the general performance of the motor	Grain number 5 of this test exhibited higher than normal mass flux behavior	Yes

Conclusion (root cause)	Over-pressure caused casing failure at the $\frac{2}{3}$ position from the bulkhead. The most likely causes were hot gas seeping through the porous grain liner (back-burn) and operating an uncharacterised propellant whose actual burn rate exceeded predictions. Poor igniter design and high nozzle-end mass flux were contributing factors.
Corrective action	• Replace ceramic cloth liner with a kraft paper tube to fully inhibit outer grain surfaces and prevent gas bypass. • Characterise the propellant empirically to determine burn rate values a and n before the next fire. • Reduction of the amount of Igniter used to avoid the initial pressure spike. • Use of surfactants inside the grain to reduce microcracks and voids.
Verification plan	• Inspect cured grain-in-tube assembly for voids or gaps. • Confirm a and n values via strand burner tests or a pressure-instrumented subscale fire. • Bench-test the new igniter in a closed chamber before full-scale use. • Clear full-scale Static Test #6 only after all three checks above pass.
Sign-off	Prepared by: _____ Reviewed by: _____ Approved by: _____ Date: _____

References

1. Solid Propulsion Team, *Theory of KNSB*, (referenced verbatim in Section 2.1, for propellant chemistry and internal-ballistics theory).
2. Solid Propulsion Team, *Revised Procedure for the Preparation of 1 Grain of KNSB Composite Sugar Propellant: Surfactant-Modified Rheology with Vacuum Degassing*, June 2026 (Sections 3.2 and 5, referenced above for grain-related evidence).

3. Solid Propulsion Team, Ballistic Evaluation Motor (BEM) Testing procedure (referenced above, Section 3.5, for static-fire verification practice).
4. 6063 aluminium alloy mechanical property data (Section 2.2), compiled from supplier and reference datasheets; T5-temper figures should be verified against the specific stock's mill certification before use as a design margin, per the caveat in that section.